

FINAL PROJECT

Isalin



A full-stack transliterator between the Latin alphabet and Baybayin — the pre-colonial Filipino syllabary — built with React, Express, and PostgreSQL.

by antoniokean

Baybayin is nearly invisible in everyday tools

— Baybayin is an abugida (syllable-based script), not an alphabet — so a simple letter-swap cipher produces wrong output.

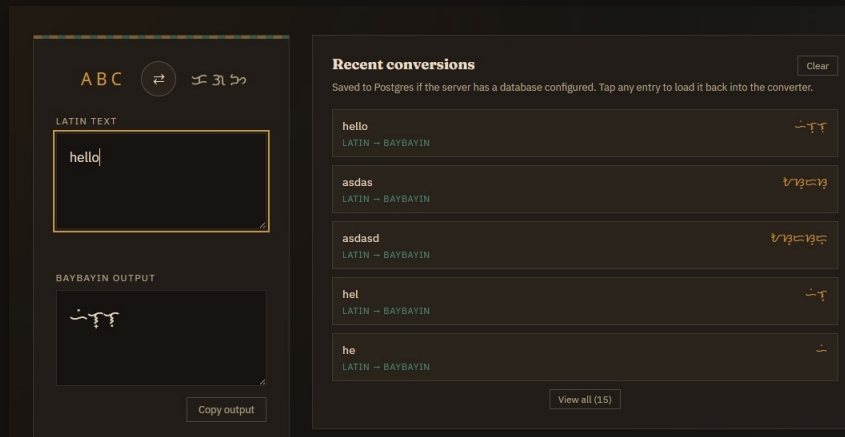
— Most “Baybayin converters” online are static image generators, not real tools that handle syllabification, kudlits, and codas correctly.

— Isalin converts how a word sounds, syllable by syllable — not a translation, a transliteration.

02 — DEMO

Type a word, get real Baybayin

- Live direction swap: Latin → Baybayin or Baybayin → Latin
- Syllable breakdown shows onset / vowel / coda for every word
- History saved to Postgres, loaded back with one click
- Password-gated and deployed publicly



The Isalin converter and its saved history

Live: isalin-deploy-6giw.vercel.app

Full stack, fully deployed

React + Vite

Express

PostgreSQL (Neon)

Render

Vercel

Unicode Tagalog block (U+1700-U+171F)

Custom syllabifier

Tokenizes words into onset + vowel + coda units before mapping to glyphs

Loan-letter substitution

For sounds Baybayin's traditional alphabet lacks (c, f, j, q, v, x, z, "ch")

Security basics

Scoped Postgres role + shared-password gate + restricted CORS

What actually went wrong

A real Unicode bug

“l”, “r”, and “w” were mapped to the wrong code points — found by tracing a misread test word back to the exact swapped hex values

Deploy misdetection

Deployment kept misdetecting the project as Express instead of Vite until the Root Directory setting was fixed

Filename casing

BreakDownView.jsx vs BreakdownView.jsx broke the build only on Vercel's Linux servers, not locally on Windows

Locked-out org repo

Couldn't deploy from the school's GitHub org repo — forking was disabled, so the project was pushed to a separate personal repo instead

Known gaps, honestly stated

— A couple of accessibility gaps remain: keyboard access on history items, one unlinked form label

— The syllabifier is a heuristic, not a full linguistic parser — handles common word shapes well, not every edge case

— No automated tests yet — verification has been manual, word by word

ALREADY SHIPPED

Security checklist fixes

Scoped DB role, password gate, restricted CORS